Physics 438B – Chapter #17
Time-Dependent Perturbation Theory

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1 Minimal Coupling

In this section, we derive the atom-field Hamiltonian in the electric-dipole approximation and motivate the need for time-dependent perturbation theory. It can be skipped if you're short on time.

In Chapter 16, we studied the quantum theory of a source-free electromagnetic field. We now want to study an atom interacting with that field. The photon operators already introduced describe the radiation field; the new question is how those operators affect the dynamics of a charged particle. It is useful to express the electromagnetic field in terms of potentials:

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \qquad \mathbf{B} = \nabla \times \mathbf{A}.$$

The potentials are not unique: different choices can produce the same electric and magnetic fields. In the *Coulomb gauge*, $\nabla \cdot \mathbf{A} = 0$, the free radiation field has $\phi = 0$, so

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}, \qquad \mathbf{B} = \nabla \times \mathbf{A}.$$

For the quantized free field in a periodic box, we can write

$$\mathbf{A}(\mathbf{r}, t) = \sum_{\mathbf{k},\mu} \sqrt{\frac{\hbar}{2V\omega_k}}\, \widehat{\mathbf{e}}_{\mathbf{k}\mu} a_{\mathbf{k}\mu} e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_k t)} + \text{h.c.}$$

This gives the electric-field operator in Chapter 16 by taking $-\partial_t \mathbf{A}$. The free-field energy is

$$H_{\text{field}} = \frac{1}{2}\int_V d^3r \left[\left(\frac{\partial \mathbf{A}}{\partial t}\right)^2 + (\nabla \times \mathbf{A})^2\right].$$

In the Coulomb gauge, once charges are present, ϕ is determined by the Poisson equation $\nabla^2\phi = -\rho$. However, we'll include the Coulomb interaction in the atomic Hamiltonian so it does not contribute another independent radiation mode.

For a single particle of charge q and mass m interacting with the radiation field, the *minimal coupling* rule gives

$$H_{\text{particle}} = \frac{[\mathbf{p} - q\mathbf{A}]^2}{2m} + q\phi + U$$

where U denotes any additional potential energy. The Hamiltonian has this form because it is invariant under a gauge transformation and it gives the classical Lorentz force equation. For hydrogen, take the proton to be fixed and use the Coulomb potential

$$U_C = -\frac{e^2}{4\pi\epsilon_0 r}$$

The total Hamiltonian describing the atom, the field, and their interaction is

$$H = H_{\text{field}} + \frac{[\mathbf{p} + e\mathbf{A}]^2}{2m_e} + U_C.$$

Include electron spin by simply adding its magnetic interaction with the field via $-\boldsymbol{\mu}_s \cdot \mathbf{B}$.

Expanding the Hamiltonian separates it into a part with no matter–field interaction and interaction terms:

$$H = H_0 + H_1 + H_2,$$

where, omitting the constant zero-point energy of the field, and using $\lambda = (\mathbf{k}, \mu)$ to label radiation modes,

$$H_0 = \frac{\mathbf{p}^2}{2m_e} + U_C(\mathbf{r}) + \sum_\lambda \hbar\omega_k \, a_\lambda^\dagger a_\lambda,$$

$$H_1 = \frac{e}{2m_e}\left[\mathbf{p}\cdot\mathbf{A}(\mathbf{r}) + \mathbf{A}(\mathbf{r})\cdot\mathbf{p}\right] + \frac{e}{m_e}\mathbf{S}\cdot\mathbf{B}(\mathbf{r}),$$

$$H_2 = \frac{e^2}{2m_e}\mathbf{A}^2(\mathbf{r}).$$

Because $\nabla\cdot\mathbf{A} = 0$, the symmetrized orbital term simplifies to $\mathbf{p}\cdot\mathbf{A} = \mathbf{A}\cdot\mathbf{p}$, so its contribution to H_1 can also be written $(e/m_e)\mathbf{p}\cdot\mathbf{A}(\mathbf{r})$. For an atom small compared with the radiation wavelength, $kr \ll 1$, expand the vector potential about the atomic center. Using the summation convention,

$$A_i(\mathbf{r}) \approx A_i(0) + r_j \partial_j A_i(0) + \cdots$$

Starting in the Coulomb gauge with $\nabla\cdot\mathbf{A} = 0$ and $\phi = 0$, define $\mathbf{A}_0(t) = \mathbf{A}(0,t)$ and choose

$$\chi(\mathbf{r}, t) = -\mathbf{r}\cdot\mathbf{A}_0(t)$$

as the "gauge function" such that under the gauge transformation

$$\mathbf{A}' = \mathbf{A} - \nabla\chi, \qquad \phi' = \phi - \partial_t\chi,$$

we have

$$\mathbf{A}' = \mathbf{A}(\mathbf{r}, t) - \mathbf{A}_0(t), \qquad \phi' = -\mathbf{r} \cdot \mathbf{E}(0, t)$$

Because $\mathbf{A}'$ is spatially constant, we are still in the Coulomb gauge, only with $\phi' \neq 0$. We can see that the transformed scalar potential gives the electric-dipole interaction

$$q\phi' = -q\,\mathbf{r} \cdot \mathbf{E}(\mathbf{0}, t) = -\mathbf{d} \cdot \mathbf{E}(\mathbf{0}, t), \qquad \mathbf{d} = q\mathbf{r}.$$

The spatial variation of $\mathbf{A}'$ contains the magnetic-dipole and higher multipole interactions. In this gauge, the constant part of $\mathbf{A}$ has been removed, leaving its spatial variation:

$$A'_i(\mathbf{r}, t) = A_i(\mathbf{r}, t) - A_i(\mathbf{0}, t) \approx r_j\, \partial_j A_i(\mathbf{0}, t).$$

Since $\mathbf{A}' = 0$, the part of the Hamiltonian that is linear in this residual vector potential is

$$H^{(1)}_{\text{orb}} = -\frac{q}{m}\, p_i r_j\, \partial_j A_i(\mathbf{0}, t).$$

Split the gradient into antisymmetric and symmetric parts:

$$\partial_j A_i = \frac{1}{2}(\partial_j A_i - \partial_i A_j) + \frac{1}{2}(\partial_j A_i + \partial_i A_j).$$

The antisymmetric part is fixed by the magnetic field:

$$\partial_j A_i - \partial_i A_j = -\epsilon_{ijk} B_k.$$

Its contribution satisfies

$$p_i r_j\, \frac{\partial_j A_i - \partial_i A_j}{2} = \frac{1}{2}\, \mathbf{L} \cdot \mathbf{B}, \qquad \mathbf{L} = \mathbf{r} \times \mathbf{p}.$$

Therefore the orbital magnetic-dipole term is

$$H_{\text{orb}} = -\frac{q}{2m}\, \mathbf{L} \cdot \mathbf{B}(\mathbf{0}) = -\boldsymbol{\mu}_L \cdot \mathbf{B}(\mathbf{0}), \qquad \boldsymbol{\mu}_L = \frac{q}{2m}\mathbf{L}.$$

Including spin adds its magnetic-moment coupling:

$$H_{\text{B-dipole}} = -(\boldsymbol{\mu}_L + \boldsymbol{\mu}_S) \cdot \mathbf{B}(\mathbf{0}).$$

For an electron with $g \simeq 2$, $\boldsymbol{\mu}_S = -e\mathbf{S}/m_e$, so

$$H_{\mathrm{B-dipole}} = \frac{e}{2m_e}(\mathbf{L} + 2\mathbf{S}) \cdot \mathbf{B}(\mathbf{0}).$$

The symmetric part of $\partial_j A_i$ gives the electric-quadrupole contribution in the multipole expansion. These terms depend on the spatial variation of $\mathbf{E}$, and in the dipole approximation, these terms are usually neglected. The only consistent way to justify the approximation is to neglect the magnetic dipole interaction as well. Indeed, the electric-dipole interaction usually dominates *when the electric-dipole transition is allowed*.

Let a be the size of the atom and $k = \omega/c$. For an allowed electric-dipole transition, the typical interaction energy is

$$H_{\mathrm{E1}} \sim e\, a\, E.$$

The orbital magnetic-dipole energy is roughly

$$H_{\mathrm{M1}} \sim \frac{e\hbar}{m_e} B \sim \frac{e\hbar}{m_e c} E,$$

using $B \sim E/c$ for radiation. Thus

$$\frac{H_{\mathrm{M1}}}{H_{\mathrm{E1}}} \sim \frac{\hbar}{m_e c a}.$$

For an atom with size $a \sim a_0$, this ratio is on the order of the fine-structure constant,

$$\frac{\hbar}{m_e c a_0} = \alpha \ll 1.$$

The quadrupole interaction is roughly

$$H_{\mathrm{E2}} \sim e\, a^2\, |\nabla E| \sim e\, a^2 k E, \qquad \frac{H_{\mathrm{E2}}}{H_{\mathrm{E1}}} \sim ka \ll 1.$$

For ordinary atomic transitions, ka is also typically of order α. So both M1 and E2 amplitudes are generally smaller than an allowed E1 amplitude. The qualification is that an E1 matrix element can vanish by selection rules. Then an M1 or E2 transition can be the leading contribution, even though it is smaller in the long-wavelength scaling. In that case, compare the allowed M1 and E2 matrix elements for the specific transition; neither automatically dominates.

We finally arrive at a fully quantum-mechanical description of the electric-dipole interaction. For hydrogen with a fixed proton, the field-free atomic Hamiltonian is

$$H_{\mathrm{atom}} = \frac{\mathbf{p}^2}{2m_e} - \frac{e^2}{4\pi\epsilon_0 r}.$$

The free radiation field is described by

$$H_{\text{field}} = \sum_{\lambda} \hbar\omega_k \, a_\lambda^\dagger a_\lambda, \qquad \lambda = (\mathbf{k}, \mu),$$

where the zero-point energy has been omitted. We take the unperturbed Hamiltonian to be

$$H_0 = H_{\text{atom}} + H_{\text{field}}.$$

Restricting attention to the electric-dipole interaction, the interaction Hamiltonian is

$$H_1 = -\mathbf{d} \cdot \mathbf{E}(\mathbf{0}),$$

where $\mathbf{d} = -e\mathbf{r}$ is the electron's dipole operator, and $\mathbf{E}(\mathbf{0})$ is the external radiation field evaluated at the atomic center. Thus, for hydrogen,

$$H_1 = e\,\mathbf{r} \cdot \mathbf{E}(\mathbf{0}).$$

Using real polarization vectors, the Schrödinger-picture field operator at the atomic center is

$$\mathbf{E}(0) = i \sum_{\mathbf{k},\mu} \sqrt{\frac{\hbar\omega_k}{2\epsilon_0 V}} \, \hat{\mathbf{e}}_{\mathbf{k}\mu} \left(a_{\mathbf{k}\mu} - a_{\mathbf{k}\mu}^\dagger \right),$$

so that

$$H_1 = ie \sum_{\mathbf{k},\mu} \sqrt{\frac{\hbar\omega_k}{2\epsilon_0 V}} \left(\mathbf{r} \cdot \hat{\mathbf{e}}_{\mathbf{k}\mu} \right) \left(a_{\mathbf{k}\mu} - a_{\mathbf{k}\mu}^\dagger \right).$$

We neglect the quadratic interaction term for the first-order electric-dipole treatment considered here. The eigenstates of H_0 are products of atomic states and photon occupation-number states:

$$|A\rangle |\{n_\lambda\}\rangle \equiv |A; \{n_\lambda\}\rangle.$$

Here $|A\rangle$ is an atomic energy eigenstate, and n_λ is the number of photons in mode λ. Their energies are

$$E_A + \sum_{\lambda} n_\lambda \hbar\omega_k.$$

The unperturbed Hamiltonian $H_0 = H_{\text{atom}} + H_{\text{field}}$ is exactly solvable: its eigenstates are products $|A; \{n_\lambda\}\rangle$ of atomic states and photon-number states. The interaction $H_1 = -\mathbf{d}\cdot\mathbf{E}(\mathbf{0})$, however, couples these states, allowing the atomic states and photon numbers to change. Expand the atom–field

state in the product eigenstates of H_0:

$$|\Psi(t)\rangle = \sum_\alpha c_\alpha(t)|\alpha\rangle, \qquad H_0|\alpha\rangle = E_\alpha|\alpha\rangle,$$

where α labels both an atomic state and a photon occupation-number state. Substitution into the Schrödinger equation gives

$$i\hbar\dot{c}_\alpha(t) = E_\alpha c_\alpha(t) + \sum_\beta \langle\alpha|H_1|\beta\rangle c_\beta(t).$$

The interaction couples each state to many others, and there are infinitely many photon modes and atomic states. Solving all these coupled equations exactly is generally impractical. Since the atom–field interaction is weak in the regime of interest, we instead solve for the coefficients approximately, treating H_1 as a perturbation. We need a *time-dependent perturbation theory.*

2 The Interaction Picture

Consider a quantum-mechanical system with a Hamiltonian

$$H(t) = H_0 + V(t) \tag{1}$$

where H_0 is a time-independent Hamiltonian that can be easily diagonalized, i.e., we know its eigenstates and corresponding energies; and V is a generally time-dependent perturbation. In the context of time-dependent perturbation theory, we wish to calculate *transition amplitudes*

$$\text{amplitude for } |i\rangle \to |f\rangle \equiv \langle f|U(t)|i\rangle \tag{2}$$

where $|i\rangle, |f\rangle$ are eigenstates of H_0 and $U(t)$ is the exact time-evolution operator (we assume $t_0 = 0$). Recall from Chapter #2 that $U(t)$ satifies a differential equation of the form

$$i\hbar\frac{\partial U(t)}{\partial t} = H(t)U(t) \tag{3}$$

In order to calculate transition amplitudes, we must solve the differential equation for $U(t)$. If $V(t)$ may be treated as a perturbation, we can use an approximation scheme.

Start by defining the "unperturbed time-evolution operator"

$$U_0(t) \equiv \exp\left(-iH_0t/\hbar\right) \tag{4}$$

Here, $U_0(t)$ describes the time evolution of the system only in the case $V(t) = 0$. Using $U_0(t)$, we can define the *interaction picture*, which is similar to the Heisenberg picture but only in terms of $U_0(t)$, not the exact time evolution operator. Think of the interaction picture as if we are factoring out the portion of the time evolution induced by H_0. Define an interaction-picture state vector in terms of the corresponding Schrödinger-picture vector as

$$|\psi_I(t)\rangle = U_0^\dagger(t)|\psi_S(t)\rangle \tag{5}$$

The vectors coincide at $t = 0$, i.e., $|\psi_I(0)\rangle = |\psi_S(0)\rangle$. For a Schrödinger picture operator A_S, define the interaction-picture operator

$$A_I = U_0^\dagger(t)A_S U_0(t) \tag{6}$$

Observe that expectation values are preserved in the two pictures, just as in the Heisenberg picture:

$$\langle\psi_I|A_I|\psi_I\rangle = \langle\psi_S|A_S|\psi_S\rangle$$

Since $|\psi_S(t)\rangle = U(t)|\psi_S(0)\rangle = U(t)|\psi_I(0)\rangle$, it follows that $|\psi_I(t)\rangle = U_0^\dagger(t)U(t)|\psi_I(0)\rangle$. Evidently, the time-evolution operator in the interaction picture is

$$W(t) \equiv U_0^\dagger(t)U(t). \tag{7}$$

Remarkably, the operator $W(t)$ satisfies a differential equation of the same form as $U(t)$ where $V_I(t)$ takes the place of the Hamiltonian. In other words,

$$i\hbar\frac{\partial W}{\partial t} = V_I(t)W(t) \tag{8}$$

where $V_I(t)$ is the interaction picture perturbation. Applying the differential equation to the initial state $|\psi_I(0)\rangle$, we find a Schrödinger equation for $|\psi_I(t)\rangle$ in the interaction picture:

$$i\hbar\frac{\partial|\psi_I(t)\rangle}{\partial t} = V_I(t)|\psi_I(t)\rangle. \tag{9}$$

The approximation scheme we employ begins with a formal integration of the differential equation for $W(t)$ from 0 to t. Integrating, we find

$$W(t) = \mathbb{1} + \frac{1}{i\hbar}\int_0^t dt'\, V_I(t')W(t'), \quad W(0) = \mathbb{1} \tag{10}$$

This expression is exact, and we notice that the portion which involves $W(t)$ on the right-hand side is proportional to the perturbation $V(t')$. Even though the expression is exact, it's an implicit

equation for $W(t)$ since it appears on both the left- and right-hand sides. We can effectively push $W(t)$ to second order in the perturbation by substituting the expression into itself... Iterating the implicit equation, we find

$$W(t) = \mathbb{1} + \frac{1}{i\hbar}\int_0^t dt'\, V_I(t') + \frac{1}{(i\hbar)^2}\int_0^t dt' \int_0^{t'} dt''\, V_I(t')V_I(t'')W(t'') \tag{11}$$

Continuing this procedure, we obtain the *Dyson series* for $W(t)$. If you like, you could introduce λ as $H = H_0 + \lambda V(t)$ to keep track of the order of each term. We can approximate the exact time evolution by truncating the series. For example, the first-order correction is

$$W^{(1)}(t) = \frac{1}{i\hbar}\int_0^t dt'\, V_I(t') \tag{12}$$

It is convenient to formulate the approximation in terms of expansion coefficients. Assume the initial state satisfies $H_0|i\rangle = E_i|i\rangle$ and $|\psi_I(0)\rangle = |i\rangle$. Using $|\psi_I(t)\rangle = W(t)|i\rangle$, we find

$$|\psi_I(t)\rangle = |i\rangle + \frac{1}{i\hbar}\int_0^t dt'\, V_I(t')|i\rangle + \frac{1}{(i\hbar)^2}(\cdots) \tag{13}$$

Expand the state vector in the unperturbed eigenstates:

$$|\psi_I(t)\rangle = \sum_n c_n(t)|n\rangle \tag{14}$$

Assuming the unperturbed eigenstates are orthonormal,

$$c_n(t) = \langle n|W(t)|i\rangle = e^{-iE_nt/\hbar}\langle n|U(t)|i\rangle \tag{15}$$

The first term represents the "rapid evolution" due to H_0 and the second term is the transition amplitude from $|i\rangle$ to $|n\rangle$. Using the approximation for $W(t)$, we can write $c_n(t)$ as a series

$$c_n(t) = \delta_{ni} + c_n^{(1)}(t) + c_n^{(2)}(t) + \cdots \tag{16}$$

where

$$c_n^{(1)}(t) = \frac{1}{i\hbar}\int_0^t dt'\, \langle n|V_I(t')|i\rangle \tag{17}$$

$$c_n^{(2)}(t) = \frac{1}{(i\hbar)^2}\int_0^t dt' \int_0^{t'} dt''\, \langle n|V_I(t')V_I(t'')|i\rangle \tag{18}$$

We can always switch back to the Schrödinger picture. For instance, the first-order correction is

$$\boxed{\begin{aligned} c_n^{(1)}(t) &= \frac{1}{i\hbar}\int_0^t dt'\,\langle n|U_0^\dagger(t')V(t')U_0(t')|i\rangle \\ &= \frac{1}{i\hbar}\int_0^t dt'\,e^{i\omega_{ni}t'}\langle n|V(t')|i\rangle \end{aligned}} \tag{19}$$

where ω_{ni} are the Bohr frequencies

$$\omega_{ni} = \frac{E_n - E_i}{\hbar} \tag{20}$$

Using a resolution of the identity, namely $\sum_k |k\rangle\langle k| = \mathbb{1}$, the second-order correction becomes

$$c_n^{(2)}(t) = \frac{1}{(i\hbar)^2}\int_0^t dt'\int_0^{t'} dt''\sum_k e^{i\omega_{nk}t'+i\omega_{ki}t''}\,\langle n|V(t')|k\rangle\,\langle k|V(t'')|i\rangle \tag{21}$$

Some people like to add colorful language to the expressions above. The first-order correction corresponds to a "direct transition" from $|i\rangle$ to $|f\rangle$, driven the the perturbation, whereas the second-order correction involves a sum over transitions through all "intermediate states" $|k\rangle$. We will mostly focus on the first-order transition amplitudes, and since we are typically interested in the transition to a state other than the one we start with, when $n = f \neq i$, the transition probability is

$$P_f(t) = |\,\langle f|U(t)|i\rangle\,|^2 = |c_f(t)|^2 \approx |c_f^{(1)}(t)|^2 \tag{22}$$

Notice that the first-order correction has the familiar form of a Fourier transform, so we interpret the result as the Fourier coefficient (in frequency space) of the perturbation at the Bohr frequency ω_{fi}. If the perturbation $V(t)$ has an appreciable component at the Bohr frequency ω_{fi} associated with the energy difference between the initial state $|i\rangle$ and some other state $|f\rangle$, then the probability of the system making a transition from $|i\rangle$ to $|f\rangle$ is large.

3 Time-Independent Perturbations

Even a time-independent perturbation can drive transitions between different unperturbed states. Taking V to be time-independent, the transition amplitude depends on the integral

$$\int_0^t dt' e^{i\omega t'} = 2e^{i\omega t/2}\,\frac{\sin(\omega t/2)}{\omega} \tag{23}$$

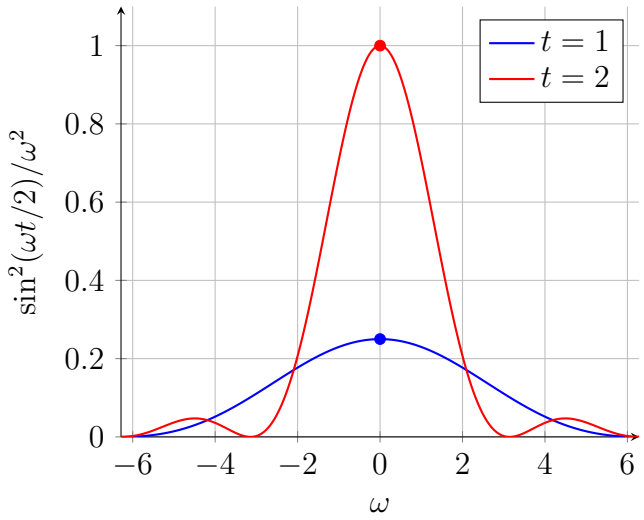


Figure 3.1: Time-dependent factor in the transition probability for $t = 1$ and $t = 2$. Longer interaction times produce a taller, narrower peak centered at $\omega = 0$.

where $\omega = \omega_{fi} = (E_f - E_i)/\hbar$. The transition probability is

$$P_f(t) = \frac{4}{\hbar^2} \frac{\sin^2(\omega t/2)}{\omega^2} \left|V_{fi}\right|^2 \tag{24}$$

where $V_{fi} = \langle f|V|i\rangle$ is the corresponding matrix element of the perturbation. The time-dependent term depends on the the unperturbed energies and the matrix element of the perturbation depends on the unperturbed states and all properties that come with them (momentum, spin, etc.)

Figure 3.1 shows the time-dependent factor in the transition probability. It is symmetric about $\omega = 0$, where its limiting value is $t^2/4$: $1/4$ for $t = 1$ and 1 for $t = 2$. As the interaction time increases, the central peak grows taller. Its first zeros are at $\omega = \pm 2\pi/t$, so they move closer to the origin as t increases. This shows that, for longer interaction times, the time-dependent factor becomes sharply concentrated around $\omega = 0$, corresponding to transitions where E_f is close to E_i.

In a distributional sense, this concentration is expressed by

$$\delta(\omega) = \lim_{t\to\infty} \frac{2}{\pi t} \frac{\sin^2(\omega t/2)}{\omega^2}. \tag{25}$$

This identity explains how energy conservation emerges in the long-time limit: transitions with nonzero energy difference contribute less, while the contribution becomes concentrated at $E_f = E_i$. For finite interaction times, transitions are possible over a range of energies around E_i. At sufficiently long times, the transition probability can become large, and first-order perturbation theory then ceases to be reliable. Defining

$$\Delta_t(\omega) = \frac{2}{\pi t} \frac{\sin^2(\omega t/2)}{\omega^2} \tag{26}$$

we can express the transition probability as

$$P_f(t) = \frac{2\pi t}{\hbar^2}\left|V_{fi}\right|^2 \Delta_t(\omega) \tag{27}$$

In the long-time limit, the linear time dependence allows us to define a **transition rate** as the probability per unit time:

$$\boxed{R_f = \frac{d}{dt}P_f(t\to\infty) = \frac{2\pi}{\hbar^2}\left|V_{fi}\right|^2 \delta(\omega)} \tag{28}$$

In many practical applications, there is a spread of final energy states rather than a single discrete quantum state. In those cases, the relevant transition rate is a sum over the rates to all accessible states. We assume that these rates are incoherent so that we can add the rates (i.e., probabilities) rather than the amplitudes.

Example 3.1: A Spin-1/2 Example

Consider

$$H_0 = \frac{\hbar\omega_0}{2}\sigma_z, \qquad V = \frac{\hbar g}{2}\sigma_x,$$

where V is constant in time. The unperturbed states are $|\uparrow\rangle$ and $|\downarrow\rangle$, with energy difference $\hbar\omega_0$. Since σ_x flips these states,

$$V_{fi} = \langle\downarrow|V|\uparrow\rangle = \frac{\hbar g}{2}.$$

If the system starts in $|\uparrow\rangle$, first-order time-dependent perturbation theory gives

$$P^{(1)}_{\text{flip}}(t) = \frac{g^2}{\omega_0^2}\sin^2\left(\frac{\omega_0 t}{2}\right).$$

Although V is constant, the transition probability depends on time. The exact result is

$$P_{\text{flip}}(t) = \frac{g^2}{\omega_0^2+g^2}\sin^2\left(\frac{\sqrt{\omega_0^2+g^2}\,t}{2}\right).$$

When $g \ll |\omega_0|$, the prefactor becomes g^2/ω_0^2, and the oscillation frequency becomes $|\omega_0|$. Thus, the perturbative result is the leading weak-coupling approximation.
In the degenerate case $\omega_0 = 0$, the exact result is

$$P_{\text{flip}}(t) = \sin^2\left(\frac{gt}{2}\right).$$

For $gt \ll 1$, this reduces to $g^2t^2/4$, the same as the perturbative result. At longer times, the exact probability oscillates, while the t^2 approximation eventually fails.

4 Time-Dependent Perturbations

The study of resonant interactions is the most important application of time-dependent perturbation theory. This is because the most efficient way to make a transition is to impose a sinusoidal perturbation *at the transition frequency*. At $t = 0$ we turn on a time-dependent perturbation Hamiltonian that has separate space and time parts:

$$V'(t) = 2V\cos(\omega_0 t) = V\left(e^{-i\omega_0 t} + e^{i\omega_0 t}\right) \tag{29}$$

where V is a Hermitian operator so that the perturbation V' is Hermitian. The probability amplitude for making a transition from $|i\rangle$ to $|f\rangle$ under the harmonic perturbation is

$$c_f(t) = \frac{2}{i\hbar}V_{fi}\left[e^{i(\omega-\omega_0)t/2}\,\frac{\sin(\omega-\omega_0)t/2}{\omega-\omega_0} + e^{i(\omega+\omega_0)t/2}\,\frac{\sin(\omega+\omega_0)t/2}{\omega+\omega_0}\right] \tag{30}$$

To find the probability, we square this amplitude, which leads to cross terms and a complicated expression. This is what we have to do if the two terms inside the square brackets are of comparable size, which happens if the frequency is far from a resonance. However, if the resonance condition is satisfied or nearly satisfied, then one of the two terms inside the square brackets dominates because the denominator approaches zero. Which term dominates depends on the sign of the energy difference.

If the final state has higher energy than the initial state, then $\omega = (E_f - E_i)/\hbar > 0$ and the first term dominates when $\omega_0 = \omega$. In this case, the dominant probability amplitude is for the transition from a lower state to an upper state, which we call **absorption**; we say that the system absorbs energy from the perturbation. If the final state has *lower* energy than the initial state, then $\omega = (E_f - E_i)/\hbar < 0$ and the second term dominates when $\omega_0 = -\omega$. The dominant probability amplitude is for the transition from a higher state to a lower state, which we call **emission**; we say that the system emits energy to the perturbation. Since the emission is "stimulated" by the perturbation, we call this **stimulated emission**.

For now, we consider the absorption case. The probability of measuring the system in the final state is

$$P_f(t) = \frac{2\pi t}{\hbar^2}\left|V_{fi}\right|^2\,\Delta_t(\omega-\omega_0) \tag{31}$$

and the corresponding transition rate in the long-time limit is given by

$$\boxed{R_f = \frac{2\pi}{\hbar^2}\left|V_{fi}\right|^2 \delta(\omega - \omega_0)} \tag{32}$$

The delta function effectively enforces energy conservation: the quantum of energy causing the transition must match the energy difference between the two states.

Before we tackle the fully quantum-mechanical problem of photon emission, let us begin with a semiclassical model. We will treat the atom quantum mechanically, but represent the electromagnetic field as a classical wave. This simpler model lets us see how an oscillating electric field drives transitions between atomic energy levels, and why the transition is strongest when the field frequency matches the energy difference between the states. We will first consider absorption, in which the atom gains energy from the field, and then stimulated emission, in which the atom transfers energy to the field. Later, we will consider the quantized field.

The interaction Hamiltonian between an atom and an applied electromagnetic field is the same as we used for the Stark effect, except that the electric field is now time dependent. We neglect the interaction of the atom with the magnetic component of the electromagnetic field because it is smaller by a factor of the fine structure constant. The electric dipole Hamiltonian is

$$V'(t) = -\mathbf{d} \cdot \mathbf{E}(t) \tag{33}$$

where $\mathbf{d}$ is the atomic dipole moment operator and the electric field is

$$\mathbf{E}(t) = 2\mathcal{E}_0 \cos(\omega_0 t)\, \widehat{\mathbf{e}} \tag{34}$$

which has the same form as the generic harmonic perturbation considered earlier. In the case of hydrogen, the perturbation can be written as

$$V'(t) = e\mathcal{E}_0\, \widehat{\mathbf{e}} \cdot \mathbf{r}\, \left(e^{-i\omega_0 t} + e^{i\omega_0 t}\right) \tag{35}$$

Since $\mathbf{r}$ is the only term in the matrix element that depends on the atomic states, we can immediately write down the transition rate as

$$R_f = \frac{2\pi e^2 \mathcal{E}_0^2}{\hbar^2} \left|\, \widehat{\mathbf{e}} \cdot\, \langle f|\mathbf{r}|i\rangle \,\right|^2 \delta(\omega - \omega_0) \tag{36}$$

The delta function is not physical, so we need to learn how to apply this to a real situation. Regardless, the transition rate clearly depends on the matrix elements of $\mathbf{r}$, and therefore obeys the

selection rules

$$\boxed{\Delta\ell = \pm 1, \quad \Delta m = 0, \pm 1} \tag{37}$$

In context, these are often called **selection rules for electric dipole transitions**. See section 14.4 of McIntyre for a derivation of these selection rules.

In order to make the situation a bit more realistic, we consider a *broadband light source* that is isotropic and unpolarized. A broadband source is composed of many frequency components. To model it, let k label a plane-wave component, including its frequency ω_k, polarization $\widehat{\mathbf{e}}_k$, and amplitude $\mathcal{E}_k$. The electric field of each component is

$$\mathbf{E}_k(t) = 2\mathcal{E}_k \widehat{\mathbf{e}}_k \cos(\omega_k t + \phi_k). \tag{38}$$

We assume the phases of different components are uncorrelated, so their transition probabilities add when averaged over time or over the source. For one component, the electric-dipole perturbation is

$$V_k'(t) = e\mathcal{E}_k \, \widehat{\mathbf{e}}_k \cdot \mathbf{r} \, \left(e^{-i\omega_k t} + e^{i\omega_k t}\right). \tag{39}$$

Its contribution to the absorption transition rate is

$$R_{i\to f}^{(k)} = \frac{2\pi e^2 \mathcal{E}_k^2}{\hbar^2} \left| \widehat{\mathbf{e}}_k \cdot \langle f|\mathbf{r}|i\rangle \right|^2 \delta(\omega - \omega_k), \tag{40}$$

We now describe how much field energy lies in a frequency interval. Recall from Phys 152/162, the time-averaged total energy density of one plane-wave component is

$$u_k = 2\epsilon_0 \mathcal{E}_k^2 \tag{41}$$

This differs from the familiar expression because we've chosen $2\mathcal{E}_k$ to be the amplitude of a single plane-wave component. The total energy density is then a sum over all plane-wave contributions

$$u = 2\epsilon_0 \sum_k E_k^2 \tag{42}$$

Now define $\rho(\omega')$ to be the time-averaged energy density per unit angular frequency interval. By definition,

$$u = \int \rho(\omega') d\omega' \tag{43}$$

For discrete components, it follows that

$$\rho(\omega') = 2\epsilon_0 \sum_k E_k^2 \delta(\omega' - \omega_k) \tag{44}$$

since integrating this expression over a frequency interval returns the energy density of the components in that interval. A real broadband source has a dense set of components, so $\rho(\omega')$ becomes a smooth spectrum.

Summing the individual transition rates gives

$$\sum_k R_{i\to f}^{(k)} = \frac{2\pi e^2}{\hbar^2} \sum_k \left| \, \widehat{\mathbf{e}}_k \cdot \langle f|\mathbf{r}|i\rangle \, \right|^2 \mathcal{E}_k^2 \delta(\omega - \omega_k) \tag{45}$$

The radiation is isotropic and unpolarized, meaning the polarization vector is randomly oriented, so we average over all possible directions of the polarization vector. The three-dimensional spatial average of $|\widehat{\mathbf{e}}_k \cdot \widehat{\mathbf{r}}|^2$ is 1/3. Therefore,

$$\begin{aligned} R_{i\to f} &= \frac{2\pi e^2}{3\hbar^2} \left| \, \langle f|\mathbf{r}|i\rangle \, \right|^2 \sum_k \mathcal{E}_k^2 \delta(\omega - \omega_k) \\ &= \frac{\pi e^2}{3\epsilon_0 \hbar^2} \left| \, \langle f|\mathbf{r}|i\rangle \, \right|^2 \rho(\omega) \end{aligned} \tag{46}$$

There are two contributions to the transition rate; there is an atomic factor determined by the states and a field factor determined by the radiation.

The atomic factor is the squared dipole matrix element. It measures how strongly the initial and final states are coupled by the position operator. If this matrix element vanishes, the transition is forbidden in the electric-dipole approximation, regardless of the intensity of the radiation. Its nonzero values give the electric-dipole selection rules, including $\Delta\ell = \pm 1$ and $\Delta m = 0, \pm 1$. The field factor, $\rho(\omega)$, is the radiation energy per unit volume per unit angular-frequency interval at the transition frequency $\omega = (E_f - E_i)/\hbar$. Only the part of the broadband radiation near this frequency contributes appreciably to the transition. Increasing the radiation energy density at ω increases the transition rate proportionally.

The atomic factor is defined as the **Einstein B coefficient** such that

$$R_{i\to f} = \frac{\pi e^2}{3\epsilon_0 \hbar^2} \left| \, \langle f|\mathbf{r}|i\rangle \, \right|^2 \rho(\omega) = B \, \rho(\omega) \tag{47}$$

The Einstein B coefficient is the same if we swap initial and final states, as is the electromagnetic energy density evaluated at the transition frequency, so the rates of emission and absorption are the same. This would suggest that a collection of two-level atoms in thermal equilibrium with a radiation field would have equal populations of its two energy levels. From statistical mechanics, we learn that the Boltzmann distribution law tells us that the populations of levels decrease as the energy level increases. There must be a third process process—**spontaneous emission**—connecting the

two levels in order for thermal equilibrium to be maintained. This is essentially the starting point for Einstein's derivation of the transition rate for stimulated emission, which we call the **Einstein *A* coefficient**. As expected, the spontaneous emission rate is independent of the applied field and is found to be

$$A = \frac{\hbar\omega^3}{\pi^2 c^3} B = \frac{e^2\omega^3}{3\pi\epsilon_0\hbar c^3} \left| \langle f|\mathbf{r}|i\rangle \right|^2 \tag{48}$$

The decay of a state caused by spontaneous emission implies that excited states are not stationary states. Excited states have an inherent finite lifetime due to spontaneous emission. If we have a system of atoms in the excited state, then the rate equation for the upper level is

$$\frac{dN_{\text{upper}}}{dt} = -N_{\text{upper}} A \tag{49}$$

Solving this differential equation yields the time dependence of the upper level population

$$N_{\text{upper}}(t) = N_{\text{upper}}(0)e^{-At} = N_{\text{upper}}(0)e^{-t/\tau} \tag{50}$$

The upper level population decays exponentially with a lifetime τ given by

$$\tau = \frac{1}{A} \tag{51}$$

We will not pursue Einstein's derivation here; rather, we will move on to calculating the Einstein A coefficient from first principles by treating both the atom and the field quantum mechanically.

5 Spontaneous Emission

Consider a single atom interacting with an electromagnetic field confined to a finite periodic box. Each quantized mode has a spatial dependence $e^{i\mathbf{k}\cdot\mathbf{r}}$, so the field variation across the atom is measured by $k\mathbf{a}$ where a is the characteristic size of the atom. If $ka \ll 1$, the field is effectively uniform over the atom and the leading interaction is the electric-dipole term. For example, for spontaneous emission of hydrogen as it transitions from $2p \to 1s$, energy conservation selects photon modes with $\omega_k = \omega_{21}$ where $\omega_{21} = (E_{2p} - E_{1s})/\hbar$. The corresponding wavenumber is $k = \omega_{21}/c$, and the relevant small parameter is

$$ka = \frac{\omega_{21} a}{c} = \frac{2\pi a}{\lambda_{21}} \tag{52}$$

For hydrogen's $2p \to 1s$ transition, $\lambda_{21} \approx 121.6\,\text{nm}$, while the orbital dimensions are on the scale of the Bohr radius. This gives $ka_{\text{atom}} \ll 1$, so the field's spatial variation across the atom is small. The dipole approximation is therefore well justified for this transition.

Take the position of the atom to be $\mathbf{r} = 0$. In the electric dipole approximation,

$$V = -\mathbf{d} \cdot \mathbf{E}(0) \tag{53}$$

In a quantization volume V, using real polarization vectors, the electric-field operator is

$$\mathbf{E}(0) = i \sum_{\lambda} \sqrt{\frac{\hbar\omega_k}{2\epsilon_0 V}}\, \widehat{\mathbf{e}}_\lambda \left(a_\lambda - a_\lambda^\dagger\right). \tag{54}$$

where $\lambda = (\mathbf{k}, \mu)$ denotes a photon mode with wavevector $\mathbf{k}$ and polarization μ. In the study of spontaneous emission, we are interested in the transition

$$|i\rangle = |2\rangle_{\text{atom}}|0\rangle_{\text{field}} \rightarrow |f\rangle = |1\rangle_{\text{atom}}|1_\lambda\rangle_{\text{field}} \tag{55}$$

We consider tensor product states of the form $|\text{atom}\rangle|\text{field}\rangle$, where the field is described in terms of photon-number states. The atom begins in a higher-energy state $|2\rangle$ while the field occupies the vacuum state. We would like to calculate the transition rate for the atom to move into the lower-energy state $|1\rangle$ with the corresponding emission of a photon in the λ mode. Remarkably, even though the field is in the vacuum state, we will find a nonzero transition rate!

Using first-order perturbation theory, we essentially have a time-independent perturbation, so the transition rate for each mode is given by Eq. (28)

$$R_f^{(\lambda)} = \frac{2\pi}{\hbar^2}\left|V_{fi}\right|^2 \delta(\omega_1 + \omega_k - \omega_2) = \frac{2\pi}{\hbar^2}\left|V_{fi}\right|^2 \delta(\omega_k - \omega_{21}) \tag{56}$$

There is no way of knowing which mode the photon will enter upon emission. The magnitude of k is constrained by energy conservation via the delta function, but the direction of propagation and polarization are not constrained. Assuming the modes are independent of each other, we can calculate the total transition rate for spontaneous emission as an incoherent sum of transition rates into each mode as

$$R_f = \frac{2\pi}{\hbar^2} \sum_{\lambda} \left|V_{fi}\right|^2 \delta(\omega_k - \omega_{21}) \tag{57}$$

Before we consider the sum over photon modes, we focus our attention on the matrix element V_{fi}. With $\mathbf{d} = -e\mathbf{r}$, the matrix elements factorize as

$$\langle f|V|i\rangle = e\,\langle 1|\mathbf{r}|2\rangle \cdot \langle 1_\lambda|\mathbf{E}|0\rangle \tag{58}$$

The annihilation operator gives zero on the vacuum, but the creation operator does not:

$$a_\lambda^\dagger|0\rangle = |1_\lambda\rangle.$$

Thus, the matrix element for emission into mode λ is, up to an irrelevant phase,

$$\langle f|V|i\rangle \;\propto\; e\sqrt{\frac{\hbar\omega_k}{2\epsilon_0 V}}\,\langle 1|\mathbf{r}|2\rangle\cdot\widehat{\mathbf{e}}_\lambda. \tag{59}$$

The squared magnitude is

$$\left|V_{fi}\right|^2 = \frac{e^2\hbar\omega_k}{2\epsilon_0 V}\left|\,\widehat{\mathbf{e}}_\lambda\cdot\,\langle 1|\mathbf{r}|2\rangle\,\right|^2 \tag{60}$$

Substituting into the transition rate, Eq. (57), we find

$$R_f = \frac{\pi e^2}{\epsilon_0\hbar V}\sum_\lambda \omega_k\left|\,\widehat{\mathbf{e}}_\lambda\cdot\,\langle 1|\mathbf{r}|2\rangle\,\right|^2\delta(\omega_k-\omega_{21}) \tag{61}$$

Using periodic boundary conditions in a cubic box with $V = L^3$, the spacing between allowed k-values is $2\pi/L$. Each wavevector therefore occupies a volume $(2\pi)^3/V$ in k-space. In a large quantization volume, the spacing between photon states becomes small, approaching a continuum as $V\to\infty$. In the continuum limit, replace sums with integrals:

$$\sum_k \to \frac{V}{(2\pi)^3}\int d^3k$$

Hence, in the continuum limit, the transition rate is independent of the quantization volume:

$$R_f = \frac{\pi e^2}{\epsilon_0\hbar(2\pi)^3}\int d^3k\sum_\mu \omega_k\left|\,\widehat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\,\langle 1|\mathbf{r}|2\rangle\,\right|^2\delta(\omega_k-\omega_{21}) \tag{62}$$

Using spherical coordinates for $\mathbf{k}$, and choosing the two real polarization vectors to be $\widehat{\boldsymbol{\theta}}$ and $\widehat{\boldsymbol{\phi}}$, we can separate the integral into a radial k-integral and an angular integral:

$$R_f = \frac{\pi e^2}{\epsilon_0\hbar(2\pi)^3}\left[\int_0^\infty k^2\,dk\,(ck)\,\delta(ck-\omega_{21})\right]\left[\int d\Omega\sum_\mu\left|\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\mathbf{r}_{12}\right|^2\right]. \tag{63}$$

where $\mathbf{r}_{12} = \langle 1|\mathbf{r}|2\rangle$ and $d\Omega = \sin\theta\,d\theta\,d\phi$. After some tedious calculations, we find

$$\int d\Omega\sum_\mu\left|\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\mathbf{r}_{12}\right|^2 = \frac{8\pi}{3}\left|\mathbf{r}_{12}\right|^2 \tag{64}$$

As for the radial integral, we use a delta-function identity to write

$$\delta(ck - \omega_{21}) = \frac{1}{c}\,\delta\left(k - \frac{\omega_{21}}{c}\right) \tag{65}$$

and thus,

$$\int_0^\infty k^2\,dk\,ck\,\delta(ck - \omega_{21}) = \int_0^\infty k^2\,dk\,ck\,\frac{1}{c}\delta\Big(k - \frac{\omega_{21}}{c}\Big) = \Big(\frac{\omega_{21}}{c}\Big)^3 \tag{66}$$

Putting the radial and angular results into the original expression,

$$\boxed{R_f = \frac{\pi e^2}{\epsilon_0 \hbar (2\pi)^3}\left(\frac{\omega_{21}^3}{c^3}\right)\left(\frac{8\pi}{3}\right)|\mathbf{r}_{12}|^2 = \frac{e^2\omega_{21}^3}{3\pi\epsilon_0\hbar c^3}\big|\,\langle 1|\mathbf{r}|2\rangle\,\big|^2.} \tag{67}$$

We have arrived at a remarkable result: *the atom can decay even when the electromagnetic field begins in its vacuum state.* The vacuum has no photons, but that does not mean that the electric-field operator vanishes. Its expectation value is zero, while its matrix element for creating a photon is nonzero. The creation-operator part of the field couples the initial state $|2;0\rangle$ to the final state $|1;1_{\mathbf{k}\mu}\rangle$. The atom loses energy $\hbar\omega_{21}$, and that energy becomes the emitted photon's energy. This is **spontaneous emission**: it does not require an incoming classical field or an initially occupied photon mode. The rate we derived is the Einstein A coefficient we discussed earlier. If we have described the atom's only decay channel, its excited-state lifetime is $\tau = 1/A$.

Consider the $2p \to 1s$ transition of hydrogen. Since there are three states in the $2p$ energy level, we must calculate the matrix element $\langle 1|\mathbf{r}|2\rangle$ for all three and then take an average. This is because we assume the atom is initially unpolarized and each of the three m_ℓ values occur with equal probability. Luckily, the matrix elements are identical, and the square magnitude is given by

$$\big|\,\langle 1|\mathbf{r}|2\rangle\,\big|^2 = \frac{2^{15}}{3^{10}}a_0^2. \tag{68}$$

Substituting this into the spontaneous emission rate gives

$$A = \frac{2^{15}}{3^{11}\pi}\frac{e^2 a_0^2 \omega_{21}^3}{\epsilon_0\hbar c^3} \tag{69}$$

Using $e^2/(4\pi\epsilon_0) = \alpha\hbar c$, this can also be written as

$$A = \frac{2^{17}}{3^{11}}\,\alpha\,\omega_{21}\left(\frac{a_0\omega_{21}}{c}\right)^2. \tag{70}$$

and since $\omega_{21} = (E_{2p} - E_{1s})/\hbar = 3\alpha c/(8a_0)$ for hydrogen,

$$A = \frac{2^8}{3^8}\,\alpha^5 \frac{m_e c^2}{\hbar} \approx 6.27 \times 10^8 \text{ s}^{-1} \tag{71}$$

This corresponds to a lifetime of

$$\tau_{2p} = \frac{1}{A} = \frac{3^8}{2^8 \alpha^5} \frac{\hbar}{m_e c^2} \approx 1.60 \times 10^{-9} \text{ s}. \tag{72}$$

So the hydrogen $2p$ lifetime is approximately 1.60 ns. The agreement with experiment is excellent, and our ability to calculate this lifetime is one of the great triumphs of quantum mechanics!